Componente Curricular: Trigonometria **Série: 2° Ano**

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1. Seja $x \in \mathbb{R}$ tal que $\cos x = \tan x$. O valor de sen x é:

a) $\frac{\sqrt{3}-1}{2}$
b) $\frac{1-\sqrt{3}}{2}$
c) $\frac{\sqrt{5}-1}{2}$
d) $\frac{1-\sqrt{5}}{2}$

$\cos x = \operatorname{tg} x$
$\cos x = \frac{\operatorname{sen} x}{\cos x}$
$\cos^2 x = \operatorname{sen} x$

$\operatorname{sen}^2 x + \cos^2 x = 1$
$\operatorname{sen}^2 x + \operatorname{sen} x = 1$
$\operatorname{sen}^2 x + \operatorname{sen} x - 1 = 0$

$a = 1$ $\quad \Delta = 1^2 - 4.1.(-1)$
$b = 1$ $\quad \Delta = 1 + 4 = 5$
$c = -1$

$\operatorname{sen} x = \frac{-1 \pm \sqrt{5}}{2}$ < $\operatorname{sen} x = \frac{-1+\sqrt{5}}{2}$; $\operatorname{sen} x = \frac{-1-\sqrt{5}}{2}$ → C.E.

2. O maior valor que o número real que $\frac{10}{2-\frac{\operatorname{sen} x}{3}}$ pode assumir é:

a) $\frac{20}{3}$
b) $\frac{7}{3}$
c) 10
d) 6
e) $\frac{20}{7}$

Para $\frac{10}{2-\frac{\operatorname{sen} x}{3}}$ ser o maior possível, $2 - \frac{\operatorname{sen} x}{3}$ deve ser o menor valor possível.
Portanto $\frac{\operatorname{sen} x}{3}$ deve ser o maior possível. Como $-1 \leq \operatorname{sen} x \leq 1$, $\operatorname{sen} x = 1$.

$\frac{10}{2-\frac{1}{3}} = \frac{10}{\frac{6}{3}-\frac{1}{3}} = \frac{10}{\frac{5}{3}} = 10 \cdot \frac{3}{5} = 6$ //

3. Se tg $\theta = 1$ e θ pertence ao primeiro quadrante, então cos θ é igual a:

a) 0
b) $\frac{1}{2}$
c) $\frac{\sqrt{2}}{2}$
d) $\frac{\sqrt{3}}{2}$
e) 1

$\operatorname{tg} \theta = 1$
$\frac{\operatorname{sen} \theta}{\cos \theta} = 1$
$\operatorname{sen} \theta = \cos \theta$

$\operatorname{sen}^2 \theta + \cos^2 \theta = 1$
$\cos^2 \theta + \cos^2 \theta = 1$
$2\cos^2 \theta = 1$
$\cos^2 \theta = \frac{1}{2}$
$\cos \theta = \pm \sqrt{\frac{1}{2}}$
$\cos \theta = \pm \frac{\sqrt{2}}{2}$
↳ 1Q → $\cos \theta = \frac{\sqrt{2}}{2}$

4. Simplificando a expressão $\frac{1+\operatorname{cotg}^2 x}{3\sec^2 x}$, onde existir, obtemos:

a) $\frac{\operatorname{tg}^2 x}{3}$
b) $3 \operatorname{cotg}^2 x$
c) $3 \operatorname{tg}^2 x$
d) $\frac{\operatorname{cotg}^2 x}{3}$
e) $\sec^2 (3x)$

$\frac{1+\frac{\cos^2 x}{\operatorname{sen}^2 x}}{3.\left(\frac{1}{\cos^2 x}\right)} = \frac{\frac{\operatorname{sen}^2 x}{\operatorname{sen}^2 x}+\frac{\cos^2 x}{\operatorname{sen}^2 x}}{\frac{3}{\cos^2 x}} = \frac{\frac{1}{\operatorname{sen}^2 x}}{\frac{3}{\cos^2 x}} = \frac{1}{\operatorname{sen}^2 x} \cdot \frac{\cos^2 x}{3} =$

$= \frac{\cos^2 x}{\operatorname{sen}^2 x} \cdot \frac{1}{3} = \operatorname{cotg}^2 x \cdot \frac{1}{3} = \frac{\operatorname{cotg}^2 x}{3}$

5. Dados tg $x = -2$ e x um arco do 2° quadrante, o valor de sec x + cossec x é:

a) $-\sqrt{5}$
b) $-\frac{\sqrt{5}}{4}$
c) $-\frac{\sqrt{5}}{2}$
d) $\frac{\sqrt{5}}{2}$
e) $\sqrt{5}$

$\operatorname{tg} x = -2$

$\frac{\operatorname{sen} x}{\cos x} = -2$

$\operatorname{sen} x = -2 \cdot \cos x$

$\operatorname{sen}^2 x + \cos^2 x = 1$

$(-2\cos x)^2 + \cos^2 x = 1$

$4\cos^2 x + \cos^2 x = 1$

$5\cos^2 x = 1$

$\cos^2 x = \frac{1}{5}$

$\cos x = \pm\sqrt{\frac{1}{5}} = \pm\frac{\sqrt{5}}{5}$

2° Q $\rightarrow \cos x = -\frac{\sqrt{5}}{5}$

$\operatorname{sen} x = -2 \cdot \left(-\frac{\sqrt{5}}{5}\right) = \frac{2\sqrt{5}}{5}$

$\sec x + \operatorname{cossec} x = \frac{1}{\cos x} + \frac{1}{\operatorname{sen} x} =$

$= -\frac{\sqrt{5}}{1} + \frac{5 \cdot \sqrt{5}}{2\sqrt{5} \cdot \sqrt{5}} = -\sqrt{5} + \frac{5\sqrt{5}}{2 \cdot 5} =$

$= -\sqrt{5} + \frac{\sqrt{5}}{2} = -\frac{2\sqrt{5}}{2} + \frac{\sqrt{5}}{2} = -\frac{\sqrt{5}}{2}$

6. Se $\begin{cases} a\cos x - \operatorname{sen} x = 1 \\ b\cos x + \operatorname{sen} x = 1 \end{cases}$, então o produto $a \cdot b$ é igual a:

a) sen$x$
b) 4
c) 2
d) 1
e) cos x

$\begin{cases} a \cdot \cos x = 1 + \operatorname{sen} x \\ b \cdot \cos x = 1 - \operatorname{sen} x \end{cases}$ ·

$(a \cdot \cos x) \cdot (b \cdot \cos x) = (1 + \operatorname{sen} x) \cdot (1 - \operatorname{sen} x)$

$a \cdot b \cdot \cos^2 x = 1 - \operatorname{sen}^2 x$

$\operatorname{sen}^2 x + \cos^2 x = 1 \rightarrow \cos^2 x = 1 - \operatorname{sen}^2 x$

$a \cdot b \cdot \cos^2 x = \cos^2 x$

$a \cdot b = \frac{\cos^2 x}{\cos^2 x}$

$a \cdot b = 1$

7. O valor da expressão $\frac{2 - \operatorname{sen}^2 x}{\cos^2 x} - \operatorname{tg}^2 x$ é:

a) −1
b) −2
c) 2
d) 1
e) 0

$\frac{2 - \operatorname{sen}^2 x}{\cos^2 x} - \frac{\operatorname{sen}^2 x}{\cos^2 x} = \frac{2 - 2\operatorname{sen}^2 x}{\cos^2 x} = \frac{2 \cdot (1 - \operatorname{sen}^2 x)}{\cos^2 x} = \frac{2 \cdot \cos^2 x}{\cos^2 x} = 2$

8. A função trigonométrica equivalente a $\frac{\sec x + \operatorname{sen} x}{\operatorname{cossec} x + \cos x}$ é:

a) sen x
b) cotg x
c) sec x
d) cossec x
e) tg x

$\frac{\frac{1}{\cos x} + \operatorname{sen} x}{\frac{1}{\operatorname{sen} x} + \cos x} = \frac{\frac{1 + \operatorname{sen} x \cdot \cos x}{\cos x}}{\frac{1 + \cos x \cdot \operatorname{sen} x}{\operatorname{sen} x}} = \frac{1 + \operatorname{sen} x \cdot \cos x}{\cos x} \cdot \frac{\operatorname{sen} x}{1 + \cos x \cdot \operatorname{sen} x} = \frac{\operatorname{sen} x}{\cos x} = \operatorname{tg} x$

GABARITO

1. C
2. D
3. C
4. D
5. C
6. D
7. C
8. E